Impact & Insight Summary

Monitoring Trends and Evaluating Intervention Impact of Lassa Fever in Nigeria (2019-2023)

1. Background and Objective

Effective public health programming requires continuous monitoring of disease trends and timely evaluation of intervention impact. This project demonstrates how routinely collected surveillance data can be used to assess disease dynamics and measure the effectiveness of public health interventions.

Using a simulated but realistic dataset reflecting Lassa fever patterns in Nigeria, the analysis evaluated trends in confirmed cases, deaths, and case fatality rate (CFR) before and after a modeled intervention introduced in early 2021.

The primary objective is to illustrate a structured, data-driven approach that supports evidence-based decision-making for public health programs and monitoring and evaluation (M&E) teams.

2. Data and Methodology

The analysis uses a five-year (2019-2023) monthly dataset covering six Nigerian states commonly associated with Lassa fever transmission. The dataset was simulated to reflect realistic epidemiological behavior, including seasonality, regional variation, and a strong intervention effect. It is intended strictly for analytical demonstration.

Key analytical steps included:

- Descriptive trend analysis of confirmed cases and deaths.
- Calculation of derived indicators, including CFR and rolling averages.
- Pre- and post-intervention comparison of average monthly outcomes.
- Visual inspection of trends with a clearly defined intervention point.

All analyses were conducted in Python using reproducible, transparent methods.

3. Key Findings

Trend Analysis: Before the intervention period, confirmed cases showed higher levels and greater volatility, particularly during early-year months consistent with known seasonal patterns. Following the intervention, a sustained decline in average monthly confirmed cases was observed across states, with a noticeable flattening of the overall trend.

Intervention Impact: A comparison of pre- and post-intervention periods indicates a substantial reduction in average monthly confirmed cases, demonstrating a strong and visible intervention effect. The decline was consistent over time rather than driven by isolated short-term fluctuations, suggesting meaningful improvement in disease control.

Mortality and Severity (CFR): Case fatality rate exhibited visible fluctuations over time, particularly during periods of lower-case counts. However, post-intervention trends show improved stability in CFR, indicating more consistent clinical outcomes alongside reduced transmission.

4. Interpretation

The findings suggest that the modeled intervention was effective in reducing disease burden and stabilizing outcomes over time. While short-term variability remained present- an expected feature of infectious disease surveillance – the overall post-intervention pattern reflects improved control rather than random variation.

Importantly, the analysis demonstrates how combining trend visualization with simple comparative metrics can provide actionable insights without relying on complex modeling. This approach is well-suited for routine program monitoring and donor reporting contexts.

5. Programmatic Implications

- Early detection and response: Trend monitoring enables timely identification of transmission shifts.
- Performance tracking: Pre- vs post-intervention comparisons provide clear evidence of program effectiveness.
- Decision support: Visual summaries and key indicators support communication with stakeholders.
- Scalability: The analytical framework can be adapted to other diseases, regions, or interventions.

6. Conclusion

This project illustrates a practical, insight-driven approach to public health program monitoring using time-series data. By integrating trend analysis, intervention comparison, and clear narrative interpretation, the framework supports evidence-based decision-making and effective communication of impact. Such approaches are essential for strengthening monitoring and evaluation practices within public health organizations.

Transparency Note

The dataset used in this project is simulated for demonstration purposes and does not represent official surveillance data. It was designed to reflect realistic patterns observed in Nigerian infectious disease monitoring.